

IP5328

Power Bank SoC with Integrated Multi-Protocol Input/Output Fast-Charge Functions

Translated from the original Chinese datasheet (Version 1.03). Circuit-diagram pin names and component labels in the original are already in English/standard notation and are reproduced as images; surrounding Chinese captions, notes, and body text have been translated.

Features

- Simultaneous support for multiple USB ports
- 2x USB-A outputs
- 1x USB-B input
- 1x USB-C input/output
- Fast-charge specifications
- Any single port supports fast charging
- Integrated QC2.0/QC3.0 output fast-charge protocol
- Integrated FCP input/output fast-charge protocol
- Integrated AFC input/output fast-charge protocol
- Integrated SFCP input/output fast-charge protocol
- Integrated MTK PE+ 1.1 & 2.0 output fast-charge protocol
- Integrated USB-C DRP protocol, supporting input/output fast charging
- Compatible with BC1.2, Apple, and Samsung fast-charge handshakes
- Charging specifications
- Battery-side charge current up to 5.0 A
- Adaptive charge-current regulation
- Supports 4.20 V, 4.35 V, 4.40 V, 4.50 V battery types
- Discharge specifications
- Output current capability: 5 V: 3.1 A, 9 V: 2.0 A, 12 V: 1.5 A
- Synchronous switching discharge, >95% efficiency at 5 V / 2 A
- Cable-drop compensation supported
- Battery-level display
- Built-in 14-bit ADC and fuel gauge
- Supports 1 / 2 / 3 / 4 LED battery-level display
- Automatically identifies the number of LED indicators installed
- Adjustable battery-level curve for more even LED transitions
- Other functions
- Built-in lighting (flashlight) LED driver
- Automatic detection of phone insertion/removal
- Fast-charge status indicator
- Key (button) support
- Smart load detection, automatically enters standby under light load
- Multiple protections, high reliability

- Input over-voltage / under-voltage protection
- Output over-current, over-voltage, short-circuit protection
- Battery over-charge, over-discharge, over-current protection
- IC over-temperature protection
- NTC protection for charge/discharge battery temperature
- ESD 4 kV; input withstand voltage 25 V
- Minimal BOM
- Integrated switching power MOSFETs
- Single inductor implements both charge and discharge functions
- Deep customization
- I2C interface for flexible, low-cost custom configurations
- Package: 6 mm x 6 mm, 0.5 mm pitch, QFN40

Overview

The IP5328 is a highly integrated power-management SoC that combines QC2.0/QC3.0 output fast-charge protocol, FCP/AFC/SFCP input/output fast-charge protocols, MTK PE+ 1.1 & 2.0 output fast-charge protocol, USB-C input/output protocol, compatibility with BC1.2/Apple/Samsung phones, a synchronous buck-boost converter, lithium-battery charge management, and battery fuel-gauge/level indication. It provides a complete power solution for fast-charge power banks.

The IP5328 can drive USB-A x2, USB-B, and USB-C simultaneously — four USB ports in total. Fast charging is supported when any single port is used alone; when two or more output ports are used at the same time, only 5 V output is supported.

Thanks to its high integration and rich feature set, the IP5328 needs only a single inductor to implement both buck and boost functions, minimizing the external component count and reducing overall BOM cost.

The IP5328's synchronous switching boost system delivers up to 18 W of output power and maintains efficiency above 90% even at low battery voltage, at full 18 W output. Under no load, the device automatically enters a sleep state.

The IP5328's synchronous switching charge system supports up to 5.0 A of charge current. Built-in IC-temperature, battery-temperature, and input-voltage control loops intelligently regulate the charge current.

The IP5328 supports 1/2/3/4 LED battery-level display with automatic recognition of the fast-charge display mode; it supports a lighting (flashlight) function and key (button) operation. Customizable functions include charge-while-discharge (pass-through) and charging wearable devices.

The IP5328 has a built-in USB-C protocol engine that supports high-current input charging and high-current output discharging via the USB-C port. A built-in 14-bit ADC precisely measures battery voltage and current, and the ADC data can be accessed via I2C. The built-in fuel-gauge algorithm accurately determines the battery's state of charge; the battery-level curve can be customized for precise battery-level indication.

Applications: power banks / battery chargers; phones, tablets, and other portable devices.

Package: 6 mm x 6 mm, 0.5 mm pitch, QFN40.

Typical Application

TYPE

Figure 1 Simplified application schematic (4-LED battery-level indication)

1. Pin Definitions

Figure 2 IP5328 pin diagram

IP5328 Pin Description

Pin No.	Pin Name	Description
1	DPA2	Fast-charge smart-identification DP for the VOUT2 port
2	CC1	USB-C detection pin CC1
3	CC2	USB-C detection pin CC2
4	DMC	Fast-charge smart-identification DM for the USB-C port
5	DPC	Fast-charge smart-identification DP for the USB-C port
6	DMB	Fast-charge smart-identification DM for the Micro-USB port; SCK when I2C2 function is enabled
7	DPB	Fast-charge smart-identification DP for the Micro-USB port; SDA when I2C2 function is enabled
8/9/22/23	VSYS	System input/output common node
10	NTC	Thermistor (NTC) detection pin
11	L1	Battery-level LED driver pin L1; SCK when I2C1 function is enabled
12	L2	Battery-level LED driver pin L2; SDA when I2C1 function is enabled
13	L3	Battery-level LED driver pin L3, multiplexed with VSET; MCU wake-up pin when I2C1 function is enabled
14/15/16/17/18	LX	DC-DC switching node; connect to the inductor
19	BST	Internal high-voltage gate-drive bootstrap; connect a capacitor to LX
20	LIGHT	Fast-charge mode status indicator
21	RSET	Battery internal-resistance compensation (fine-tunes battery-level curve); MCU wake-up pin when I2C2 function is enabled
24	VSP	VSYS current-sense positive terminal; must be routed separately from VSYS
25	VSN	VSYS current-sense negative terminal
26	KEY	Key (button) detection pin, multiplexed with the WLED lighting function
27	VREG	Chip 3.1 V regulator output
28	BAT	Battery supply node
29	AGND	Analog ground
30	VIN	VIN charging power-source detection pin
31	VING	VIN charging-input PMOS control pin
32	VBUS	VBUS charging power-source detection pin
33	VBUSG	VBUS charging-input PMOS control pin
34	VOUT2	VOUT2 discharge-load smart-detection pin
35	VOUT2G	VOUT2 discharge-output NMOS control pin
36	VOUT1G	VOUT1 discharge-output NMOS control pin
37	VOUT1	VOUT1 discharge-load smart-detection pin
38	DMA1	Fast-charge smart-identification DM for the VOUT1 port

Pin No.	Pin Name	Description
39	DPA1	Fast-charge smart-identification DP for the VOUT1 port
40	DMA2	Fast-charge smart-identification DM for the VOUT2 port
41 (EPAD)	GND	Power ground / thermal ground; must maintain good contact with GND

2. IP-Series Model Selection Guide

Power-Bank ICs

Model	Discharge	Charge	LED Count	Light	Key	I2C	DCP	USB C	QC Cert.	Package
IP5303	1.0A	1.2A	1,2	√	√	-	-	-	-	eSOP8
IP5305	1.0A	1.2A	1,2,3,4	√	√	-	-	-	-	eSOP8
IP5306	2.4A	2.1A	1,2,3,4	√	√	-	-	-	-	eSOP8
IP5206	2A(Max)	1.5A	3,4,5	√	√	-	-	-	-	eSOP16
IP5108E	2.0A	1.0A	3,4,5	√	√	-	-	-	-	eSOP16
IP5108	2.0A	2.0A	3,4,5	√	√	√	-	-	-	eSOP16
IP5207	1.2A	1.2A	3,4,5	√	√	-	-	-	-	QFN24
IP5207T	1.2A	1.2A	1,2,3,4	√	√	√	√	-	-	QFN24
IP5109	2.1A	2.1A	3,4,5	√	√	√	-	-	-	QFN24
IP5209	2.4A	2.1A	3,4,5	√	√	√	√	-	-	QFN24
IP5219	2.4A	2.1A	1,2,3,4	√	√	√	√	√	-	QFN24
IP5310	3.1A	3.0A	1,2,3,4	√	√	√	√	√	-	QFN32
IP5312	15W	3.6A	2,3,4,5	√	√	√	√	-	-	QFN32
IP5318Q	18W	4.0A	2,3,4,5	√	√	√	√	-	√	QFN40
IP5318	18W	4.0A	2,3,4,5	√	√	√	√	√	√	QFN40
IP5322	18W	4.0A	1,2,3,4	√	√	√	√	-	√	QFN32
IP5328	18W	4.0A	1,2,3,4	√	√	√	√	√	√	QFN40

Protocol ICs

Note: the source PDF's multi-row header for this table did not extract cleanly; column order below is reconstructed from context and may not match the manufacturer's exact header wording.

Model	Ch.	BC1.2/ Apple	QC2.0	QC3.0	FCP	SCP	AFC	SFCP	MTK PE+1.1 &2.0	USB C	NT C	PD3.0 Cert.	Package
IP2110	1	√	-	-	-	-	-	-	-	-	-	-	SOT23-5
IP2111	1	√	-	-	-	-	-	-	-	-	-	-	SOT23-6
IP2112	2	√	-	-	-	-	-	-	-	-	-	-	SOT23-6
IP2161	1	√	√	√	-	√	√	-	-	√	-	-	SOT23-6
IP2163	1	√	√	√	-	√	√	√	-	√	√	-	SOP8
IP2701	1	√	√	√	-	√	√	-	√	-	-	-	SOP8
IP2703	1	√	√	√	-	√	√	√	√	√	-	-	DFN1

Model	Ch .	BC1.2/ Apple	QC2. 0	QC3. 0	FCP	SCP	AF C	SFC P	MTK PE+1.1 &2.0	USB C	NT C	PD3. 0 Cert.	Packa ge
													0
IP2705	1	√	√	√	-	√	√	√	√	√	-	-	DFN1 2
IP2707	2	√	√	√	-	√	√	√	√	√	-	-	QFN1 6
IP2716	1	√	√	√	√	√	-	1.1	√	-	√	√	QFN3 2

3. Absolute Maximum Ratings

Parameter	Symbol	Value	Unit
Port input-voltage range	VIN, VBUS	-0.3 ~ 16	V
Junction-temperature range	TJ	-40 ~ 150	°C
Storage-temperature range	Tstg	-60 ~ 150	°C
Thermal resistance (junction to ambient)	θJA	26	°C/W
Human Body Model (HBM)	ESD	4	kV

* Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device. Prolonged exposure to conditions beyond the Absolute Maximum Ratings may affect device reliability and lifetime.

4. Recommended Operating Conditions

Parameter	Symbol	Min	Typ	Max	Unit
Input voltage	VIN, VBUS	4.5	5	14	V
Battery voltage	Vbat	3.0	3.7	4.5	V

* Device operating characteristics are not guaranteed outside these operating conditions.

5. Electrical Characteristics

Unless otherwise specified, TA = 25°C, L = 2.2μH.

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
Charging system						
Input voltage	VIN, VBUS		4.5	5/7/9/12	13	V
Input over-voltage threshold	VIN, VBUS		13	14	15	V
Charge target (constant) voltage	VTRGT	RVSET = NC	4.16	4.2	4.24	V
		RVSET = 120kΩ	4.31	4.35	4.39	V
		RVSET = 68kΩ	4.36	4.4	4.44	V
		RVSET = 10kΩ	4.46	4.5	4.54	V
Charge current	ICHRG	VIN = 5V, input current		2.0		A
		VBUS = 5V, input current		2.8		A
		VIN or VBUS ≥ 7V, input power		18		W
Trickle-charge current	ITRKL	VIN = 5V, BAT < 1.5V	50	100	150	mA
		VIN = 5V, 1.5V ≤ BAT < 3.0V	100	250	400	mA
Trickle-charge cutoff voltage	VTRKL		2.9	3	3.1	V
Charge-termination current	ISTOP		200	300	400	mA

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
Recharge threshold voltage	VRCH		4.08	4.1	4.13	V
Charge timeout	TEND		20	24	27	Hour
Boost (discharge) system						
Battery operating voltage	VBAT		3.0		4.5	V
Switching quiescent battery input current	IBAT	VBAT = 3.7V, VOUT = 5.1V, fs = 375kHz		3	5	mA
DC output voltage	VOUT	QC2.0: VOUT = 5V @ 1A	4.95	5.12	5.23	V
		QC2.0: VOUT = 9V @ 1A	8.75	9	9.25	V
		QC2.0: VOUT = 12V @ 1A	11.75	12	12.25	V
		QC3.0: @ 1A	4.95		12.25	V
		QC3.0: step size		200		mV
Output voltage ripple	ΔV_{OUT}	VBAT = 3.7V, VOUT = 5.0V, fs = 375kHz		100		mV
Boost-system output current	Iout	VOUT = 5V			3.1	A
		VOUT = 9V			2.0	A
		VOUT = 12V			1.5	A
Boost-system efficiency	η_{out}	VBAT = 3V, VOUT = 5V, IOUT = 2A		95		%
		VBAT = 3V, VOUT = 9V, IOUT = 2A		92		%
		VBAT = 3V, VOUT = 12V, IOUT = 1.5A		90.8		%
Boost over-current shutdown threshold	Ishut	VBAT = 3.7V, output sense resistor 10m Ω	3.5	3.8	4.0	A
Load over-current detection time	TUVD	Output voltage continuously below 4.2V		30		ms
Load short-circuit detection time	TOCD	Output current continuously above 4.2A	150	200		us
Control system						
Switching frequency	fs	Discharge switching frequency	325	375	425	kHz
		Charge switching frequency	450	500	550	kHz
NMOS on-resistance	rDSOn	High-side FET		9	11	m Ω
		Low-side FET		9	11	m Ω
VREG output voltage	VREG	VBAT = 3.7V	3.0	3.1	3.2	V
Battery-side standby current	ISTB	VIN = 0V, VBAT = 3.7V, average current		100		μ A
LDO output current	ILDO		20	30	40	mA
LED lighting-driver	IWLED		10	15	20	mA

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
current						
LED display-driver current	IL1, IL2, IL3	Voltage drop 10%	5	7	9	mA
Total-load light-load auto-shutdown detection time	T1load	Total load power continuously < 300mW	25	32	44	s
Per-port light-load auto-shutdown detection time	T2load	VSN-to-output voltage difference continuously < 1.8mV		T1load / 2		s
Short key-press wake time	TOnDebounce		60		500	ms
WLED turn-on (long-press) time	TKeylight		1.2	2	3	s
Thermal-shutdown temperature	TOTP	Rising temperature	130	140	150	°C
Thermal-shutdown hysteresis	ΔTOTP			40		°C

6. Functional Description

Under-voltage Lockout and Activation

The first time a battery is connected, the IP5328 always starts in a locked state regardless of the battery voltage; the lowest-order battery-level LED blinks 4 times as an indication. Likewise, if the battery voltage becomes too low while not charging and triggers a low-voltage shutdown, the IP5328 re-enters the locked state.

In the locked state, to minimize quiescent power the IP5328 does not perform phone-insertion detection and cannot be activated by the key (button). Pressing the key in this state cannot activate the boost output, but the lowest-order battery-level LED will blink 4 times as an indication.

In the locked state, the chip must enter the charging state before its functions can be activated.

Charging

The IP5328 has a synchronous-switching constant-current/constant-voltage lithium-battery charge-management system that automatically matches different charge-voltage specifications.

When the battery voltage is below 3 V, the charger uses 250 mA trickle charging. When the battery voltage exceeds 3 V, the charger enters input constant-current mode, with a maximum battery-side charge current of 5.0 A. As the battery voltage approaches the configured target voltage, the charger enters constant-voltage mode; once the battery-side charge current falls to roughly below 300 mA and the battery voltage is near the constant-voltage target, charging stops. After charging completes, if the battery voltage subsequently drops below 4.1 V, charging restarts automatically.

The IP5328 uses switching charge technology at a 500 kHz switching frequency. With a standard 5 V input, maximum charge input power is 10 W; with a fast-charge input, maximum charge input power is 18 W. Battery-side charge current can reach up to 5.0 A, with charging efficiency up to 94%, shortening charge time by up to 3/4.

The IP5328 automatically adjusts the charge current to match adapters of different current capability.

The IP5328 does not support simultaneous charge and discharge (pass-through) in its standard configuration — discharge output ports are disabled while charging, to protect the device being charged from the higher input voltage.

Boost (Discharge)

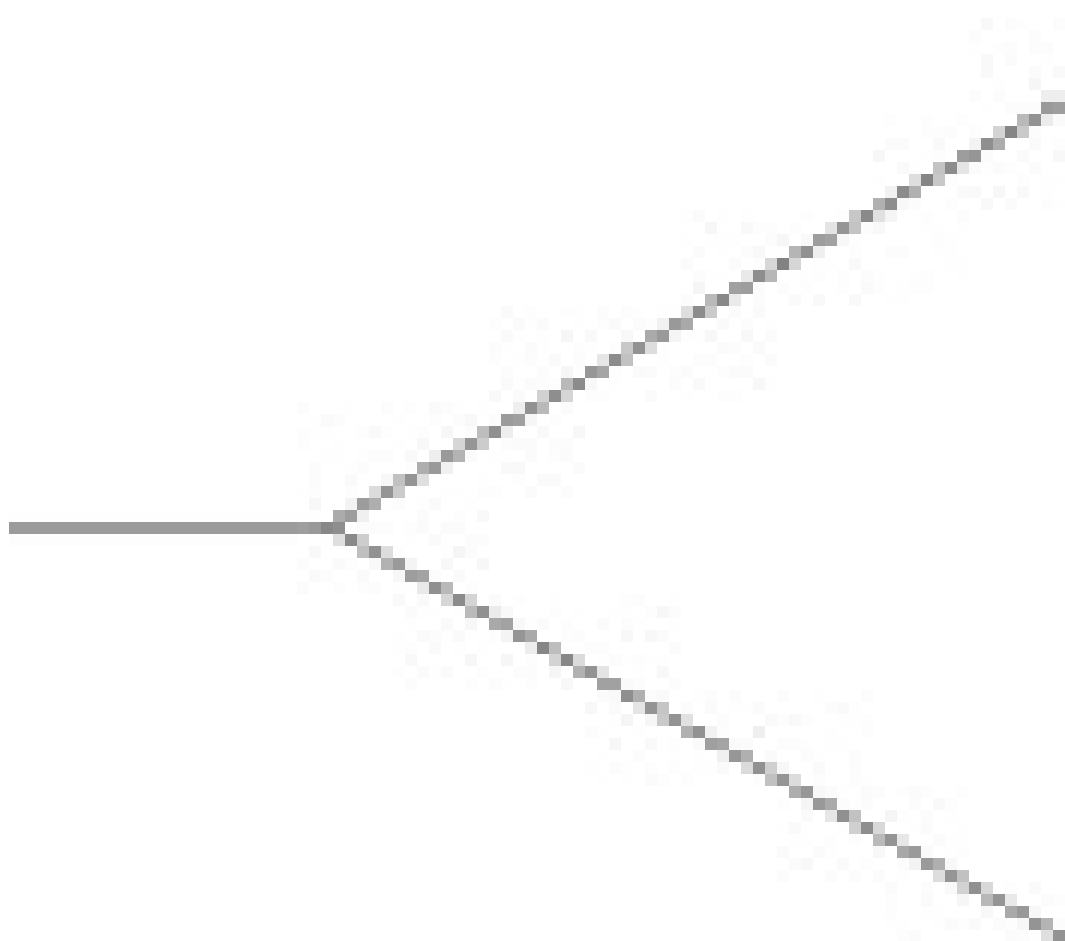
The IP5328 integrates a synchronous switching boost converter system supporting a wide 5 V – 12 V output range, with load capability of 5V—3.1A, 7V—2.4A, 9V—2.0A, and 12V—1.5A, at a 375 kHz switching frequency. A built-in soft-start function prevents excessive inrush current at startup, and integrated output over-current, short-circuit, over-voltage, and over-temperature protections ensure stable, reliable system operation.

The boost system's output current is automatically derated with temperature to keep the IC temperature below the configured limit.

USB C

The IP5328 integrates a USB-C DRP input/output recognition interface that automatically switches its internal pull-up/pull-down resistors to auto-detect the charge/discharge role of the connected device. It supports the Try.SRC function, which preferentially sources power to the far-end device when it is also a DRP.

When acting as a DFP, the IP5328 can advertise one of three output current-capability levels externally; when acting as a UFP, it can identify the far-end device's output current capability.



USB-C CC pull-up / pull-down network

Pull-up / pull-down capability

Name	Value
Ip_3P0	330 μ A
Rd	5.1 k Ω

Comparator threshold when pull-up current source Ip is enabled:

Comparator threshold when pull-down resistor Rd is enabled:

USB-C Detection Cycle

The device cycles between advertising as a Source and as a Sink per the USB Power Delivery / Type-C DRP timing specification (parameters below reproduced from the USB-C specification, already in English in the source datasheet).

USB-C Detection State Transition

(No state-transition diagram was present at this location in the source PDF.)

Fast-Charge Function

The IP5328 supports multiple fast-charge specifications: QC2.0/QC3.0, FCP, AFC, SFCP, MTK, Apple, and Samsung.

Charging the power bank itself does not use QC — if QC input charging is required, a Qualcomm QC charger IC (e.g., SMB1351) can be added on the charging input port.

Charging the power bank can support FCP, AFC, and SFCP fast-charge input. Because FCP, AFC, and SFCP negotiate their fast-charge handshake over DP/DM, if another fast-charge protocol IC is added on the input, FCP/AFC/SFCP input fast charging can no longer be supported.

When the power bank charges a phone: after entering discharge mode, the IP5328 automatically detects the fast-charge handshake timing on the DP/DM pins and intelligently identifies the phone type. It supports QC2.0/QC3.0, FCP, AFC, SFCP, and MTK protocol phones, as well as Apple phones in 2.4 A mode, Samsung phones in 2 A mode, and BC1.2 standard Android phones in 1 A mode.

For Apple-phone support: DP = DM = 2.7 V

For Samsung-phone support: DP = DM = 1.2 V

For BC1.2 support: DP shorted to DM

In BC1.2 mode, when the DP voltage is detected to be greater than 0.325 V and less than 2 V for 1.25 s continuously, this is a preliminary indication of a fast-charge request; the DP–DM short is then opened and DM is pulled down through 20 k Ω to ground. If, for 2 ms continuously, DP remains between 0.325 V and 2 V while DM is below 0.325 V, the fast-charge connection is considered successful, after which the requested voltage can be output per the QC2.0/QC3.0 protocol. Whenever DP drops below 0.325 V, the device forces an exit from fast-charge mode and the output voltage immediately returns to the default 5 V.

QC2.0 / QC3.0 Output Voltage Request Rules

DP	DM	Result
0.6V	GND	5V
3.3V	0.6V	9V
0.6V	0.6V	12V
0.6V	3.3V	Continuous Mode
3.3V	3.3V	Hold

Continuous Mode is a QC3.0-specific operating mode in which the output voltage can be finely adjusted in 0.2 V steps per the QC3.0 protocol.

Charge / Discharge Path Management

Standby

If a charging source is plugged into VIN or VBUS, charging starts directly.

If a USB-C UFP device is plugged into VBUS, or a load device is plugged into VOUT, discharge is enabled automatically.

If the key is pressed, the VOUT1 port is forced on regardless of whether a load is present; VOUT2 or the USB-C port only turns on if a load is present, otherwise it remains off. Therefore, if a design uses only one VOUT port, VOUT1 must be retained; if VOUT2 or VBUS is unused, its MOSFET can be omitted, but the 10 μ F capacitor must still be kept.

Discharging

Without a key press, only the output path of a port with a connected load device is opened; the output path of an unconnected port is not opened. Once opened, a port's output path automatically closes after its output current stays below roughly 180 mA @ 10 m Ω (i.e., the voltage difference between VSN and the output stays below 1.8 mV; if

the impedance between VSN and the output — including MOSFET impedance — exceeds 10 m Ω , the current threshold scales down proportionally) for a period of time.

When only VOUT1 output is enabled, a single short key press has no effect. When VOUT2 or USB-C output is enabled and not in fast-charge mode, a single short key press forces VOUT1 on. When VOUT2 or USB-C output is enabled and in fast-charge mode, the first short key press turns off fast charging on the VOUT2/USB-C output, and the second short key press forces VOUT1 on; the interval between the two short presses must exceed 1 s, otherwise a forced shutdown is triggered.

Any of VOUT1, VOUT2, or USB-C can support all output fast-charge protocols, but because this is a single-inductor design, only one output voltage can be supported at a time — fast-charge output is only available when a single output port is active. When two or three output ports are used simultaneously, fast charging is automatically disabled.

Per the “typical application schematic” wiring, if one output port is already in fast-charge mode and a load device is plugged into another output port (a key press is equivalent to a load being plugged into VOUT1), all output ports are first turned off and high-voltage fast charging is disabled; then the port with the connected device is re-enabled, but at this point all output ports support only Apple/Samsung/BC1.2 charging. When the number of connected load devices drops from multiple down to just one, after roughly 16 s all output ports are first turned off (this behavior depends heavily on correct PCB layout — otherwise automatic fast-charge recovery may fail; see the Layout Notes section), high-voltage fast charging is re-enabled, and finally the port with the one remaining device is turned back on, which re-triggers that device's fast-charge request. When only one output port is enabled and its total output power stays below roughly 300 mW for about 32 s, the output port and discharge function are turned off and the device enters standby.

Charging

Either the VIN or VBUS port can be used to charge the device; if a source is connected to both at once, the source that was plugged in first is used preferentially.

If a design adds an external input fast-charge protocol IC, the input port with that added IC must use two back-to-back PMOS devices for isolation, to prevent the high voltage on the fast-charge input port from backfeeding into the standard charging port when both are connected simultaneously.

In single-charger mode, the fast-charge mode of the source is automatically identified, and the appropriate charge voltage and current are automatically matched.

The standard configuration does not support charge-while-discharge — even if a device is plugged into an output port while charging, that output port is not enabled. However, once the power bank is fully charged and the charging source has not been unplugged, the IP5328 will enable an output port to charge a connected device whenever the input voVoltage is 5 V (i.e., not in an input fast-charge state).

Charge-While-Discharge (Custom Order Only)

In the charge-while-discharge custom configuration, the RSET pin no longer provides the standard-configuration RSET function; instead it outputs low when not in charge-while-discharge mode and high when in charge-while-discharge mode.

When a charging source and a load device are connected at the same time, the device automatically enters charge-while-discharge mode. In this mode, the chip automatically disables input fast charging (for designs with an external input fast-charge protocol IC, the RSET pin is used to control that IC's enable signal). If VSYS is at 5 V, the discharge path is enabled to power the load device; if VSYS exceeds 5.8 V, the discharge path is not enabled, for safety. To ensure the load device charges normally, the IP5328 raises the charge under-voltage loop threshold above 4.9 V, to prioritize supplying the load device.

During charge-while-discharge, if the charging source is unplugged, the IP5328 disables charging and restarts the discharge function to power the load device. For safety, and to allow the phone to re-request fast charging, the voltage briefly drops to 0 V during this transition.

During charge-while-discharge, if the load device is unplugged, or stops drawing current once fully charged, the corresponding discharge path automatically turns off after about 16 s. Once all discharge paths are closed and the device returns to single-charge mode, the charge under-voltage loop threshold is lowered again, automatically re-enabling fast charging to speed up charging the power bank.

Automatic Phone Detection

Phone-insertion auto-detection

The IP5328 automatically detects phone insertion and immediately wakes from standby, turning on 5 V boost output to charge the phone — no key press is required, enabling button-less enclosure designs.

Phone fully-charged auto-detection

The IP5328 senses output current through an external 10 m Ω resistor. When total output power stays below roughly 300 mW for about 32 s, all connected phones are considered fully charged or removed, and boost output is automatically disabled.

Key (Button)

Figure 3 Key (button) wiring

The key is wired as shown in Figure 3, and both long-press and short-press actions are recognized.

- A press lasting more than 60 ms but less than 2 s is a short press, which turns on the battery-level LEDs and boost output.
- A press lasting more than 2 s is a long press, which toggles the WLED lighting function on or off.
- A press shorter than 30 ms produces no response.
- Two short presses within 1 s turn off boost output, the battery-level display, and the WLED light.
- An extra-long 10 s press resets the entire system.

Fast-Charge Status Indicator

The LIGHT pin indicates the current fast-charge mode. Whether charging or discharging, whenever the device enters fast-charge mode at a non-5V voltage, the indicator LED automatically turns on.

⇒ 肌肉内无快肌纤维

2. 

LIGHT indicator LED

Fuel Gauge and Battery-Level Display

The IP5328 has a built-in fuel-gauge function that computes accurate battery state-of-charge without directly sensing BAT current.

The IP5328 flexibly supports one-, two-, three-, or four-LED battery-level display configurations. A built-in smart-recognition algorithm automatically identifies how many external battery-level LEDs are populated.

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Figure 4 Pin connections for 4-LED, 3-LED, 2-LED, and 1-LED display modes

Diagram labels (translated): 4 灯模式 = 4-LED mode; 3 灯模式 = 3-LED mode; 1 灯模式1 = 1-LED mode 1; 2 灯模式1 = 2-LED mode 1; 2 灯模式2 = 2-LED mode 2.

4-LED Display Behavior

While charging:

Battery level C (%)	D1	D2	D3	D4
Full	On	On	On	On
$75\% \leq C$	On	On	On	1.5 Hz blink
$50\% \leq C < 75\%$	On	On	1.5 Hz blink	Off
$25\% \leq C < 50\%$	On	1.5 Hz blink	Off	Off
$C < 25\%$	1.5 Hz blink	Off	Off	Off

While discharging:

Battery level C (%)	D1	D2	D3	D4
$C \geq 75\%$	On	On	On	On
$50\% \leq C < 75\%$	On	On	On	Off
$25\% \leq C < 50\%$	On	On	Off	Off
$3\% \leq C < 25\%$	On	Off	Off	Off
$0\% < C < 3\%$	1.0 Hz blink	Off	Off	Off
$C = 0\%$	Off	Off	Off	Off

3-LED Display Behavior

While charging:

Battery level C (%)	D1	D2	D3
Full	On	On	On
$66\% \leq C$	On	On	1.5 Hz blink
$33\% \leq C < 66\%$	On	1.5 Hz blink	Off
$C < 25\%$	1.5 Hz blink	Off	Off

While discharging:

Battery level C (%)	D1	D2	D3
$C \geq 66\%$	On	On	On
$33\% \leq C < 66\%$	On	On	Off
$3\% \leq C < 33\%$	On	Off	Off
$0\% < C < 3\%$	1.0 Hz blink	Off	Off
$C = 0\%$	Off	Off	Off

2-LED Mode 1 (Bi-color LED)

While charging:

Battery level C (%)	D1	D2
Full	Off	On
$66\% \leq C < 100\%$	Off	1.5 Hz blink
$33\% \leq C < 66\%$	1.5 Hz blink	1.5 Hz blink
$C < 33\%$	1.5 Hz blink	Off

While discharging:

Battery level C (%)	D1	D2
$66\% \leq C < 100\%$	Off	On
$33\% \leq C < 66\%$	On	On
$C < 33\%$	On	Off
$C < 3\%$	1.0 Hz blink	Off

2-LED Mode 2

While charging: D1 blinks with a 2.0 s period (1.0 s on / 1.0 s off); when full, it stays on continuously.

While discharging: D2 stays on continuously; below 3.2 V it blinks at 1.0 Hz (0.5 s on / 0.5 s off); below 3.0 V the device shuts down.

1-LED Mode 1

While charging: blinks with a 2.0 s period (1.0 s on / 1.0 s off); when full, stays on continuously.

While discharging: stays on continuously; below 3.2 V it blinks at 1.0 Hz (0.5 s on / 0.5 s off); below 3.0 V the device shuts down.

RSET (Battery Internal-Resistance Setting)

The IP5328 can set the modeled battery internal resistance via the RSET pin, which also shifts the charge/discharge LED-transition threshold voltages, allowing the LED battery-level display to be tuned for more even transitions. The relationship between the RSET resistor value and the configured battery internal resistance is shown below. (This function is unavailable in the charge-while-discharge custom configuration, where internal resistance is fixed at 50 m Ω .)

RSET resistance (k Ω)	Configured battery internal resistance (m Ω)
> 179	93.75
169 ~ 179	87.5
159 ~ 169	81.25
149 ~ 159	75
139 ~ 149	68.75
129 ~ 139	62.5
119 ~ 129	56.25
109 ~ 119	50
99 ~ 109	43.75
89 ~ 99	37.5
79 ~ 89	31.25
69 ~ 79	25
59 ~ 69	18.75
49 ~ 59	12.5
39 ~ 49	6.25
0 ~ 39	0

VSET (Battery Type Setting)

The IP5328 can set the battery type via the VSET pin, which changes the battery-level display thresholds, the charge constant-voltage target, and protection voltages. The relationship between the VSET resistor value and the configured battery type is shown below.

VSET resistance (k Ω)	Battery type
NC	4.2V
120	4.35V
68	4.4V
10	4.5V

NTC Function

The IP5328 integrates an NTC function to sense battery temperature. During operation, the NTC pin sources a 20 μ A current and the voltage on the NTC pin is measured to determine the current battery temperature.

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$$V = 20$$

$$R1 = 82K$$

Figure 5 Battery NTC comparator

Formula: $V(\text{NTC}) = 20\mu\text{A} \times (R1 // R_{\text{NTC}})$, with $R1 = 82 \text{ k}\Omega$ and $R_{\text{NTC}} = 100 \text{ k}\Omega @ 25^\circ\text{C}$, $B = 4100$.

While charging: the NTC pin detecting 1.32 V represents 0°C (battery too cold) — charging stops.

The NTC pin detecting 0.56 V represents 45°C (battery warm) — charge current is reduced by half.

The NTC pin detecting 0.49 V represents 50°C (battery hot) — charging stops.

While discharging: the NTC pin detecting 1.44 V represents -15°C (battery too cold) — discharge stops.

The NTC pin detecting 0.43 V represents 55°C (battery hot) — discharge stops.

If a design does not require NTC, a $51 \text{ k}\Omega$ resistor must be tied from the NTC pin to ground — it must not be left floating or tied directly to ground.

VREG

VREG is an always-on 3.1 V LDO with 30 mA load capability.

I2C

I2C Connection Modes

RSET

I2C Mode 2 (left) and I2C Mode 1 (right)

The chip can support only one I2C wiring configuration at a time; connecting the pins per the corresponding mode automatically disables the Function-pin behavior and enters I2C mode. When wired for I2C mode, the INT signal is high-impedance in standby and driven high during operation, and can be used to wake the MCU.

I2C mode supports up to 400 kbps, with 8-bit register addresses and 8-bit register data; both transmit and receive are MSB-first. I2C device address: 0xEA for write, 0xEB for read.

Example: writing data 0x5A to register 0x05

SCLK

SDA

I2C Write

Example: reading data back from register 0x05

从0x05寄存器

SCLK

SDA

SI

Start

I2C Read

In I2C mode, RSET is low when the IP5328 is powered off and high when the IP5328 is powered on, so RSET can be used as an MCU wake-up control signal.

7. Layout Notes

Only a few notes that may affect function and performance are listed here; additional notes will be supplied in a separate document if needed.

VSYS Capacitor Placement

The chip operates at relatively high power and current, so the placement of capacitors on the VSYS net affects DC-DC stability. VSYS-net capacitors should be placed as close as possible to the IC's VSYS pins and EPAD, with a large copper pour and additional vias, to minimize the loop area — and thus parasitics — between the capacitors and the IC.

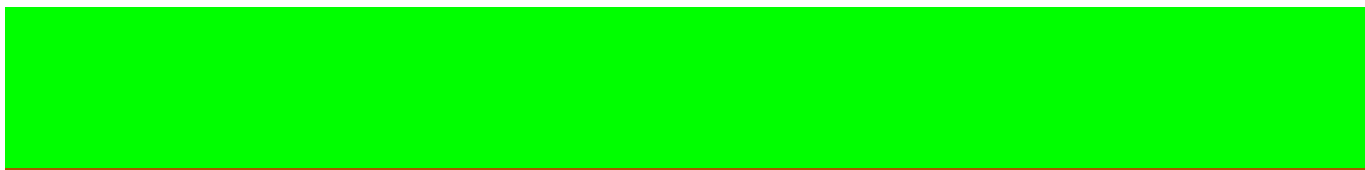
VSYS pins are distributed on both sides of the chip; capacitors should be placed close to the pins on each side, and the VSYS pins on both sides should be tied together on the PCB using a wide copper pour (no narrower than 100 mil).

10 mΩ Sense Resistor

The chip senses the current through the 10 mΩ resistor via the VSN and VSP pins to implement input charge-current sensing, output over-current protection, and output light-load auto-shutdown. Consequently, the PCB routing requirements for the VSN and VSP pins are demanding: these traces must avoid noisy signals and must be routed individually to the inside of the two 10 mΩ sense-resistor pads, without any overlap with VSYS or any other current-carrying path. Although VSP and VSYS are the same net on the PCB, their pin traces must be routed completely separately.

The VSN and VSP pins each require a 100 nF filter capacitor, placed as close to the IC pins as possible to improve the sensed signal's noise immunity.

As shown below:



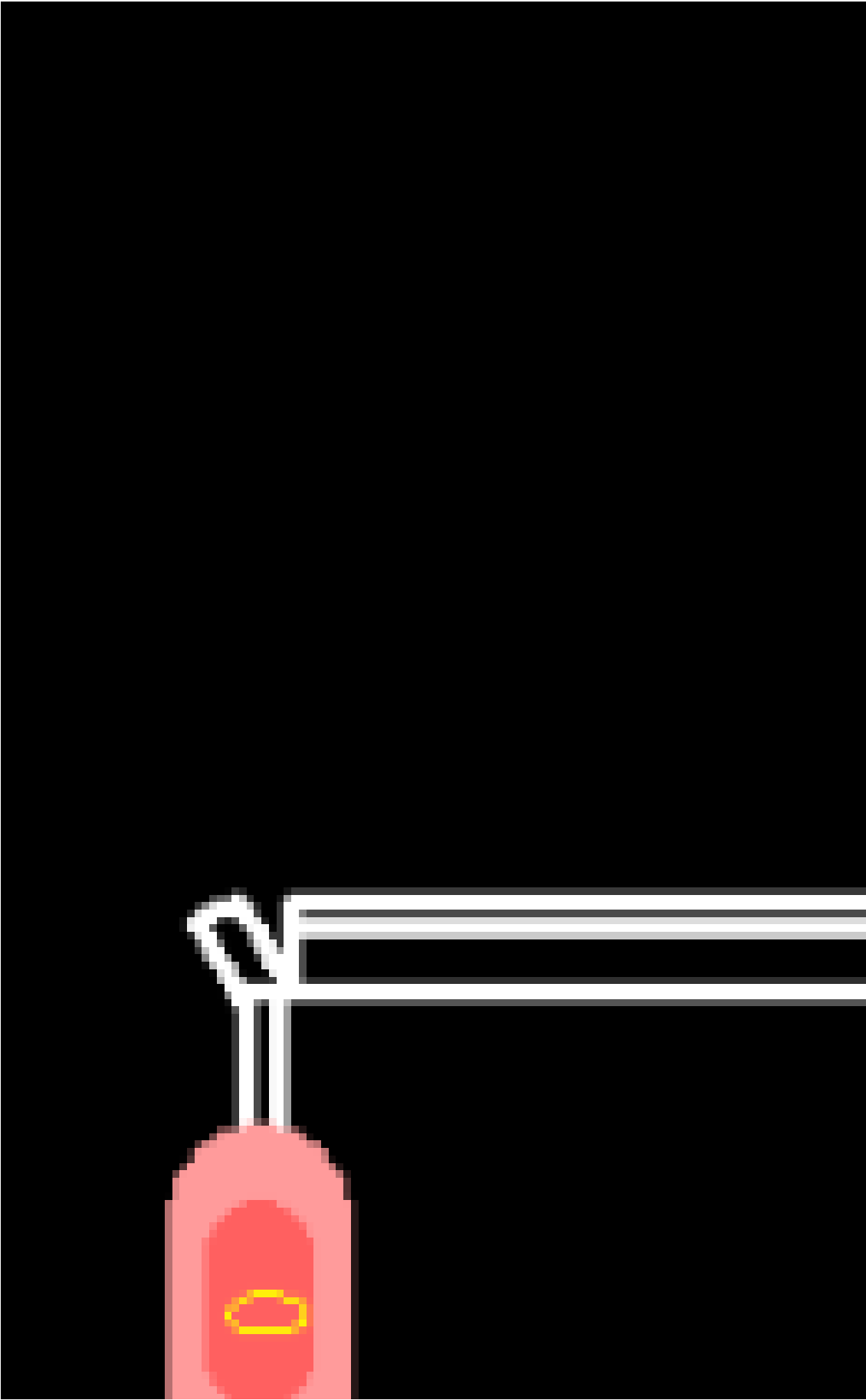
Routing from the 10 mΩ Sense Resistor to the Input/Output MOSFETs

Example application: a non-fast-charge phone is plugged into VOUT1 and a fast-charge-capable phone is plugged into VOUT2. Because both outputs are active simultaneously, the system can only output 5 V to both phones. When the phone on VOUT1 is unplugged, or its current draw drops below the configured threshold, the system turns off the VOUT1 output; with only VOUT2 remaining active, the system can then automatically restore fast charging on VOUT2 for the connected phone.

To achieve this automatic fast-charge recovery, the output current on VOUT1 must be accurately sensed. VOUT1's output current is sensed from the voltage difference between the VSN and VOUT1 pins; the threshold for closing the VOUT1 port is a voltage difference below 1.8 mV. Therefore, when there is no current on VOUT1, no other current may flow through any segment of trace between VSN and VOUT1 — any such current would create a voltage difference and be misread as current flowing out of VOUT1. The same principle applies to the other output ports.

In summary, when routing the PCB, the three traces from the 10 mΩ resistor to VOUT1, from the 10 mΩ resistor to VOUT2, and from the 10 mΩ resistor to VIN must be separated right at the 10 mΩ resistor; no two traces carrying current in the same direction may overlap, or the automatic fast-charge-recovery behavior described in the example above may fail intermittently.

As shown below:



Routing from the sense resistor to the output MOSFETs

If incorrect layout causes a $1\text{ m}\Omega$ length of overlapping trace between the VOUT1 and VOUT2 output currents, then when VOUT2 is sourcing 2 A, the overlapping $1\text{ m}\Omega$ segment produces a 2 mV voltage difference. In this situation, even unplugging the device on VOUT1 cannot be detected, so VOUT2's fast-charge output cannot be restored. The VOUT2 output current must first drop below 1.8 A, so that the voltage difference produced across the overlapping $1\text{ m}\Omega$ trace falls below 1.8 mV, for 16 ms continuously, before VOUT2's fast-charge output can automatically recover.

8. Typical Application Schematic

The IP5328 requires only MOSFETs, an inductor, capacitors, resistors, and a few other passive components to implement a full-featured fast-charge power-bank solution.

Configuration: TYPE-A + TYPE-A + Micro-B + USB-C

Typical application schematic (TYPE-A + TYPE-A + Micro-B + USB-C)

Diagram notes (translated): the four traces from R1 to VOUT1, R1 to VOUT2, R1 to VBUS, and R1 to VIN must separate right at the R1 pad, with no overlap in their current paths, otherwise automatic single-port fast-charge recovery when switching from multiple ports to one port will be affected. When laying out, keep power-path traces as short and wide as possible — trace width should be at least 80 mil. When the NTC function is unused, replace R12 with 51 k Ω and leave RNTC unpopulated. Block labels: WLED lighting LED; fast-charge status indicator (LIGHT); key (button); output port 2 / output port 1 (support 5V/3.0A, 7V/2.4A, 9V/2.0A, 12V/1.5A output; VOUT2 connector/Q1 NMOS/CP9 can be omitted if VOUT2 is unused, but CP9 must be retained; VOUT1 must be retained if only one VOUT is used); charging input port (supports 5V/7V/9V/12V input charging); USB-C port (supports 5V/3.0A, 7V/2.4A, 9V/2.0A, 12V/1.5A input/output; the USB-C connector, Q2 PMOS, and R8 can be omitted if USB-C is unused, but CP10 must be retained).

BOM

No.	Component	Type & Spec	Ref. Des.	Qty	Notes
1	SMD IC	QFN40, IP5328	U1	1	
2	SMD capacitor	0603, 100nF, 10%, 25V	C3, C4	3	
3	SMD capacitor	0603, 2.2 μ F, 10%, 16V	C1, C2	2	
4	SMD capacitor	0805, 22 μ F, 10%, 16V	CP1, CP2, CP3, CP14	4	
5	SMD capacitor	0805, 22 μ F, 10%, 25V	CP4, CP5, CP6, CP8	4	
6	SMD capacitor	0805, 10 μ F, 10%, 25V	CP7, CP9, CP10, CP13	4	
7	Electrolytic capacitor	220 μ F, 25V, 10%	CP12	1	
8	SMD resistor	1206, 0.01 Ω , 1%	R1	1	
9	SMD resistor	0603, 20 Ω , 5%	R4	1	Adjusts LED brightness
10	SMD resistor	0603, 100 Ω , 5%	R2, R5, R7	3	
11	SMD resistor	0603, 1k Ω , 5%	R6	1	
12	SMD resistor	0603, 10k Ω , 5%	R8, R9	2	
13	SMD resistor	0603, 110k Ω , 1%	R10	1	
14	SMD resistor	0603, 82k Ω , 1%	R12	1	Required for the NTC circuit
15	NTC thermistor	100k Ω @ 25°C, B = 4200	RNTC	1	Required for the NTC circuit
16	SMD LED	0603, blue	D1, D2, D3, D4	4	
17	SMD LED	0603, green	D8	1	
18	LED	5mm LED	D5	1	
19	Molded inductor	2.2 μ H, 10x10mm	L1	1	
20	SMD NMOS	SOT23-3, RU207C-I	Q1, Q3	2	Rds(on) < 20m Ω , I $\geq$ 4A
21	SMD PMOS	SOT23-3, RU20P7C-I	Q2, Q4	2	Rds(on) < 20m Ω , I $\geq$ 4A
22	USB-C receptacle	USB-C receptacle	J1	1	
23	Button	SMT 3x6 button	SW1	1	

No.	Component	Type & Spec	Ref. Des.	Qty	Notes
24	Output USB	AF10, 8-pin USB connector	USB1, USB3	2	
25	Input USB	MICRO-7-DIP-5.9	USB2	1	

Recommended Inductors

Vendor P/N	Thickness (mm)	Inductance (μH)	Tolerance	DC Resistance Typ. (mΩ)	DC Resistance Max. (mΩ)	Heat Rating Current Idc(A) Max.	Saturation Current Isat(A) Max.	Measuring Condition
SPM70702R2M ESQ	5	2.2	±20%	9	10.2	10.5	13.5	100kHz / 1.0V
SPM10102R2M ESN	4	2.2	±20%	6	7	12	18	100kHz / 1.0V
SHC1004-2R2M	4	2.2	±20%	7	9	12	24	

9. Package Information



QFN40, 6mm x 6mm, 0.5mm pitch

Symbol	Min (mm)	Nom (mm)	Max (mm)
A	0.70	0.75	0.80
A1	-	0.035	0.05
A2	-	0.65	0.67
A3	-	0.125	-
b	0.2	0.25	0.30
e	0.5 BSC		
D	6 BSC		
E	6 BSC		
J	4.52	4.62	4.72
K	4.52	4.62	4.72
L	0.35	0.40	0.45
aaa	0.1		
bbb	0.1		
ccc	0.08		
ddd	0.1		
eee	0.1		